

1st Semester Physics Topics for preparation (MID)

PHYSICS-1 (PHY-1101)		MID Term Exam topics_Autumn 2026
1st Chapter: Dynamics of Rigid Body	2nd Chapter: Gravity and Gravitation	3rd Chapter: Introductory Quantum Mechanics
Definition type (CLO1): Rigid body, Moment of Inertia, Center of Mass, Torque, Collision, Conservation theorem of energy (Only statement), Conservation theorem of momentum (Only Statement)	Definition type (CLO1): Newton's Law of Gravitation, Gravitational field, Gravitational Potential, Gravitational Potential Energy, Kepler's Laws of Motion: 1 st law, 2 nd law, 3 rd law (statements only), Escape Velocity	Definition type (CLO1): Wave function, the probability in Quantum Mechanics, the expectation value, Heisenberg's Uncertainty Principle
Broad type (Proper theory, explanation, calculation, figure) CLO2: 1) Rotational kinetic energy 2) Perpendicular & parallel axis theorem of moment of inertia 3) Law of conservation of momentum of two bodies in a bound system.	Broad type (Proper theory, explanation, calculation, figure) CLO2: 1) Gravitational Potential due to a Spherical Shell: (a) at a point outside the shell , (b) at a point inside the shell . 2) Derive an equation for the time period of a compound pendulum	Broad type (Proper theory, explanation, calculation, figure) CLO2: 1) Schrödinger Time Independent Equation Or, Steady State form of the Schrodinger equation 2) Time Dependent Schrödinger Equation

Maths From Chapter 1

1) **Scenario:** In 1985, Test Devices (Incorporation) was spin-testing a sample of a solid steel rotor (a disk) of mass $M = 272 \text{ kg}$ and radius $R = 38.0 \text{ cm}$. When the sample reached an angular speed ω of $14,000 \text{ rev/min}$, how much energy was released in the explosion of the rotor?

2) A thin spherical shell has a radius of 1.90 m . An applied torque of 960 Nm gives the shell an angular acceleration of 6.20 rad/s^2 about an axis through the center of the shell. What are (a) the rotational inertia of the shell about that axis and (b) the mass of the shell? [For a thin spherical shell, moment of inertia, $I = \frac{2}{3} MR^2$]

3) The blade of a small wind turbine is rotating about an axis of rotation, and the center of mass is at a distance of 2.5 m from that axis of rotation. The total mass of the rotating rigid body is 10 kg . Find the moment of inertia.

4) A flywheel of mass 100 kg and radius 0.5 m makes 10 revolutions per second. Calculate (i) angular velocity, (ii) moment of inertia, and (iii) the energy of the flywheel. A flywheel is treated as a solid uniform disc.

5) A rigid body is rotating about the Z-axis of rotation, and the center of mass is at a distance of 4 cm from that axis of rotation. The total mass of the object is 5 kg. Find (i) the moment of inertia, I_z , (ii) If $I_x = 0.005 \text{ kg-m}^2$, find the value of I_y .

Maths From Chapter 2

1) If an object of mass 5 kg is situated at a distance of 200 cm from the center of a sphere of radius 110 cm, determine the value of (i) the gravitational potential and (ii) the gravitational potential energy (GPE). [The sphere has a mass of 100 kg]

2) If a body is projected in such a way that it escapes the Earth's gravitational field. Then find the required velocity of that body. [$G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$, $M = 6 \times 10^{24} \text{ kg}$, $R = 6 \times 10^6 \text{ m}$]

3) What is the gravitational potential with respect to the Sun at the position of the Earth? The mass of the Sun is 1.99×10^{30} kilograms, and the mass of the Earth is 5.98×10^{24} kilograms. The mean Earth-Sun distance is 150×10^6 kilometers.

4) Calculate the gravitational force of attraction between two metal spheres, each of mass 90 kg, if the distance between their centres is 10^3 m . Given $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$. Will the force of attraction be different if the same bodies are taken to the Moon, with their separation remaining the same?

5) Two particles of masses 1.0 kg and 2.0 kg are placed at a separation of 50 cm. Assuming that the only forces acting on the particles are their mutual gravitation, find the initial accelerations of the two particles.

6) Calculate the escape velocity from the Moon. The mass of the Moon = $7.4 \times 10^{22} \text{ kg}$ and the radius of the Moon = 1740 km.

7) Find the distance of a point from the Earth's centre where the resultant gravitational field due to the Earth and the Moon is zero. The mass of the earth is $6.0 \times 10^{24} \text{ kg}$, and that of the Moon is $7.4 \times 10^{22} \text{ kg}$. The distance between the earth and the Moon is $4.0 \times 10^5 \text{ km}$.

8) Four particles having masses m , $2m$, $3m$ and $4m$ are placed at the four corners of a square of edge a . Find the gravitational force acting on a particle of mass m placed at the centre.

9) Find the acceleration due to gravity of the moon at a point 1000 km above the moon's surface. The mass of the moon is $7.4 \times 10^{22} \text{ kg}$, and its radius is 1740 km.

Maths From Chapter 3

1) Why can't an electron be present in the nucleus? Explain the reason with the uncertainty principle.

2) An electron is confined to a box of length 10^{-10} m . Calculate the minimum uncertainty in its velocity.

3) A proton is confined to a box of length $5 \times 10^{-10} \text{ m}$. Calculate the minimum uncertainty in its momentum and kinetic energy of the proton.

4) An electron has a speed of 600 m/s with an accuracy of 0.005%. Calculate the certainty with which we can locate the position of the electron. Comment on your answer.

5) A bullet of mass 0.025 kg is moving with a speed of 400 m/s. The speed is measured accurate to 0.02%. Calculate the certainty with which we can locate the position of the bullet.

[Note: You may need to put the values of simple parameters such as the gravitational force constant, mass of Earth, mass of Sun, mass of the proton, the electron, and Max Planck's constant yourself. They may not be provided in the exam]